

Chapter 8

Angles

In this chapter we will study an important property of inversive geometry known as **the preservation of angles**. More precisely, in the upcoming trio of theorems we will prove that under an inversion of a plane with respect to a circle the magnitude of the angles between the original straight lines and between their inversive images remains the same.

Definition 4. The angle between two curves at a point, T , is the angle between the tangents to both curves at that point, T .

For the sake of simplicity and in general, we will initially consider only those curves that are at least once differentiable at the points of interest because such curves admit the existence of the limiting position of the respective secant or a tangent (straight line).

From the **Definition 4** it follows that once we prove that an inversion of a plane with respect to a circle preserves the magnitudes of the angles between straight lines, then, pretty much automatically, that would imply that such an inversion also preserves the magnitudes of the angles between a straight line and a circle and that the said inversion also preserves the magnitudes of the angles between two circles.

As such.

Theorem 32. Under the inversion of a plane with respect to a circle, $c(O, r)$, the magnitude of the angle between any two straight lines, l_1 and l_2 , both of which pass through the center of inversion, O , is equal to the magnitude of the angle between their inversive images, l'_1 and l'_2 .

Proof. Let l_1 and l_2 be two straight lines, each of which passes through the center of inversion. Then, according to the Theorem 9, each such straight line will be transformed into itself. Therefore, the magnitude of the angle between the straight lines l_1 and l_2 will be equal to the magnitude of the angle between their inversive images. $\square$

Note that the Theorem 32 applies to inversions with both positive and negative powers.

Theorem 33. Under the inversion of a plane with respect to a circle, $c(O, r)$, the magnitude of the angle between any two straight lines, l_1 and l_2 , of which exactly one passes through the center of inversion, O , is equal to the magnitude of the angle between their inverse images, l'_1 and l'_2 .

Proof. Without the loss of generality let us assume that the straight line l_1 passes through the center of inversion, O , while the straight line l_2 does not.

From the Theorem 9 it follows that the inverse image, l'_1 , of the straight line l_1 with respect to the circle of inversion, c , will also pass through the center of inversion, O , and will coincide with the preimage straight line l_1 .

From the Theorems 10, 11, 12, 13 and 14 it follows, then, that under the inversion of the parent plane with respect to a circle, c , with both positive and negative powers the remaining straight line l_2 will be transformed into a circle l'_2 that passes through the center of inversion, O .

From the Definition 4 it follows that the angle between the images l'_1 and l'_2 will be the angle between the straight line l'_1 and the tangent straight lines, t and t' , drawn to the circle l'_2 at certain points, see below.

Looking at an inversion with **positive** power, we consider three cases:

- the straight lines l_1 and l_2 are parallel
- the straight lines l_1 and l_2 are perpendicular
- the straight lines l_1 and l_2 are neither parallel nor perpendicular

Case 1. The straight line l_1 and l_2 are parallel.

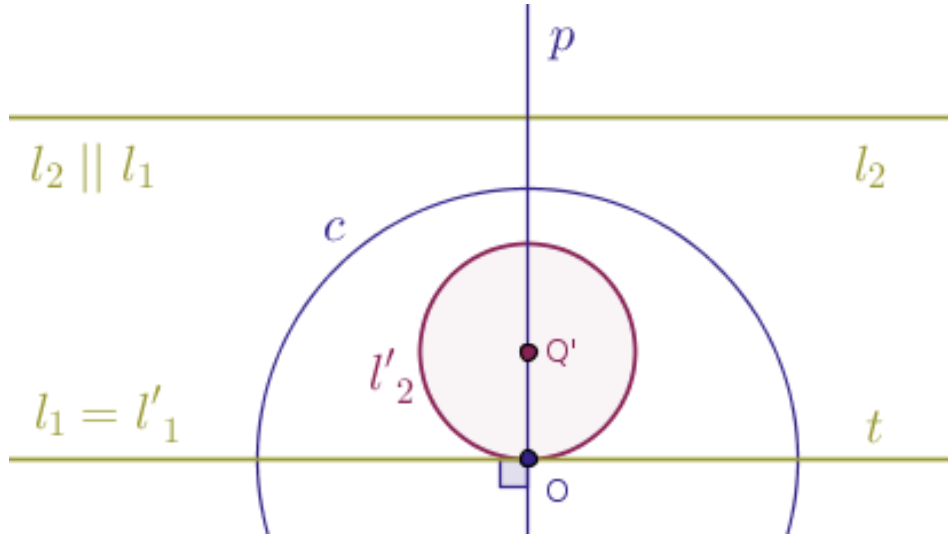
When the straight lines l_1 and l_2 are parallel then the angle between them is zero radians.

By construction of the circle l'_2 , our straight line of centers, p , the straight line that passes through the center of inversion, O , is perpendicular to the preimage straight line l_2 . By the definition of the tangent straight line, t , drawn to the circle l'_2 at the point O , the straight line p will also be perpendicular to the straight line t .

We, thus, have a straight line, p , falling on two straight lines, l_2 and t , making the alternate angles equal one another by magnitude. Therefore, by Euclid's "Elements" Book 1 Proposition 27, the tangent straight line t is parallel to the preimage straight line l_2 (we shall reuse this argument in the remaining cases).

But if the straight line t is parallel to the straight line l_2 and that straight line, l_2 , by hypothesis, is parallel to the straight line l_1 , it follows that, by [Euclid's "Elements" Book 1 Proposition 30](#), t is parallel to l_1 .

We, thus, have two parallel straight lines, l'_1 and t , passing through one and the same point O , which means that, by [the Playfair's axiom](#), these two straight lines must coincide, which means that the magnitude of the angle between these two straight lines must also be zero radians (Figure 8.1):



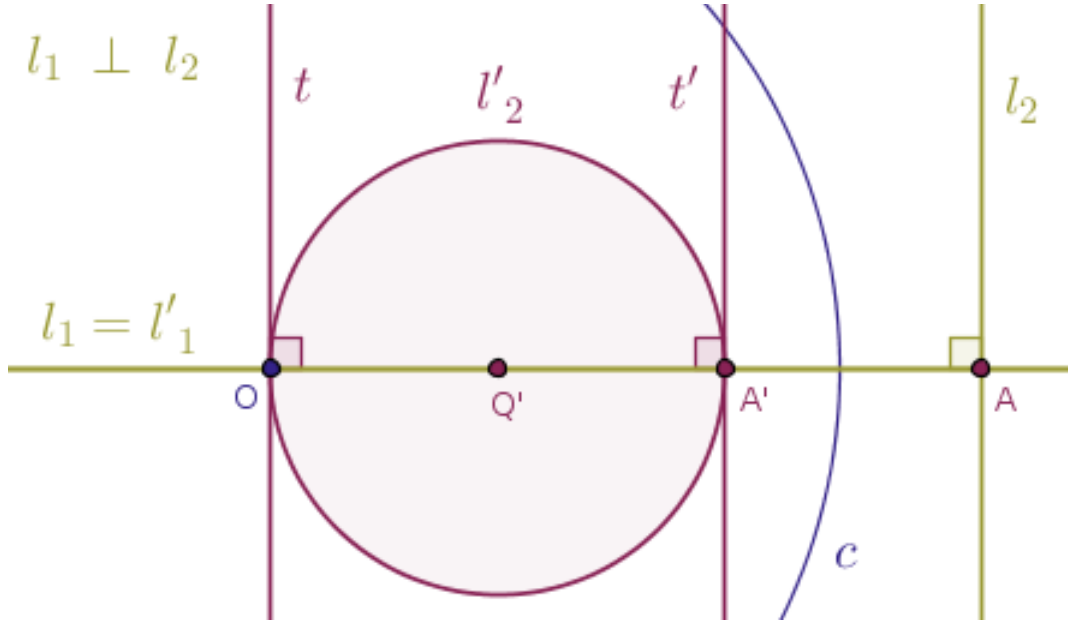
Thus, in this case the magnitude of the angles between the original straight lines, l_1 and l_2 , and between their inverse images, l'_1 and l'_2 is preserved. $\square$

Case 2. The straight lines l_1 and l_2 are perpendicular.

When the straight lines l_1 and l_2 are perpendicular then by the parallel postulate these two straight lines will intersect. Let l_1 and l_2 intersect at a point A . Then, the magnitude of the angle between l_1 and l_2 at the point A will be $\pi/2$ radians.

Let the point A' be the inverse of the point A and let l'_2 be the inverse of the straight line l_2 under the inversion of a plane with respect to a circle c with positive power.

Then, the points O , A , A' and Q' (the center of the image circle l'_2), by the Theorems 10, 11, 12, 13 and 14, will all be located on the straight line l'_1 . Hence, the diameter $OQ'A'$ of the circle l'_2 will also be located on the straight line l'_1 as well (Figure 8.2):



By [Euclid's "Elements" Book 1 Proposition 27](#) (please see **Case 1** above), both tangent straight lines to the circle l'_2 , t at O and t' at A' , will be parallel, since the corresponding angles between the diameter $OQ'A'$ and t and t' are right by definition (of a tangent to a circle at a point).

By the same Book 1 Proposition 27, the tangent straight line t will be parallel to l_2 and the tangent straight line t' will be parallel to l_2 , which makes all three straight lines t , t' , l_2 parallel.

Because the above straight lines are parallel, by [Euclid's "Elements" Book 1 Proposition 29](#), a straight line, $l_1 = l'_1$, falling on two parallel straight lines, t and t' , makes the magnitudes of the alternate angles equal to one another; it follows that t and t' are perpendicular to l'_1 which, thus, proves that the magnitude of the angles between the original straight lines l_1 and l_2 and between their inverse images is also preserved. $\square$

Case 3. The straight lines l_1 and l_2 are neither parallel nor perpendicular.

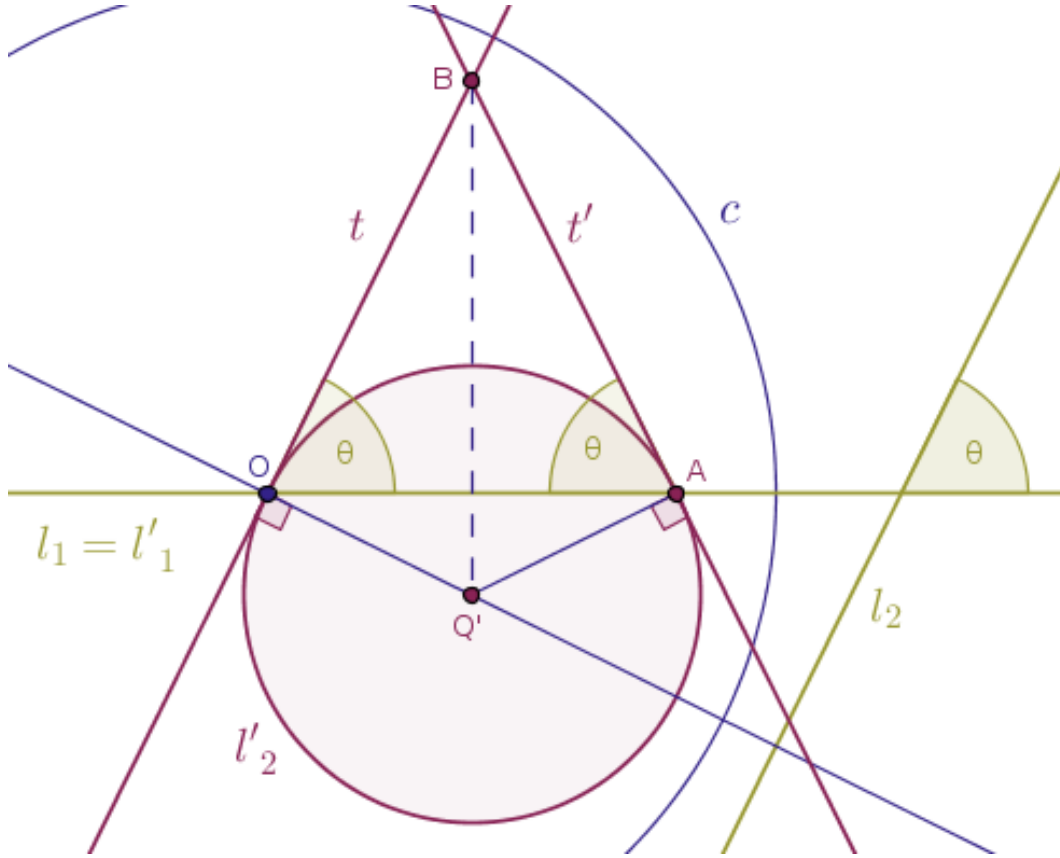
When the straight lines l_1 and l_2 are neither parallel nor perpendicular then by the same parallel postulate these two straight lines will intersect. We, then, take the angle θ between l_1 and l_2 to be the smallest one.

Moreover, let the tangent straight line t to the circle l'_2 at the point O and the tangent straight line t' to l'_2 at the point A intersect at a point B .

In this case, again, by the same Book 1 Proposition 27, the tangent straight line t to l'_2 at the point O will be parallel to the original straight line l_2 . But then the straight line l'_1 will be falling on two parallel straight lines t and l_2 and, thus, by [Book 1 Proposition 29](#), it must be the case that:

$$\angle BOA = \theta$$

which means that the magnitude of the angles between the original straight lines l_1 and l_2 and between their inverse images l'_1 and l'_2 at the point O is preserved (Figure 8.3):



Next, consider the angle BAO between the second tangent straight line t' and l'_1 at the point A . By construction and by the definition of a tangent straight line, the angles $Q'AB$, where Q' is the center of the image circle l'_2 , and $Q'OB$ are right and $|Q'O| = |Q'A|$, since both line segments are the radii of one and the same circle l'_2 .

Because the hypotenuse BQ' is shared across the right triangles $Q'AB$ and $Q'OB$, by the Pythagorean theorem or by [Euclid's "Elements" Book 1 Proposition 47](#), it follows that $|BO| = |BA|$ also. Therefore, the triangle ABO is isosceles and due to the famous Pons Asinorum or [Book 1 Proposition 5](#), the magnitudes of the interior angles at the base OA of that triangle must be equal: $\angle BAO = \angle BOA = \theta$.

which means that the magnitude of the angles between the original straight lines l_1 and l_2 and between their inverse images l'_1 and l'_2 at the point A is preserved as well. $\square$

Our readers are encouraged to generate the relevant proof of the Theorem 33 as it applies to an inversion of a plane with negative power on their own. Here the utility of the Theorem 8, where we proved that an inversion of a plane with negative power is equivalent to the inversion of the same plane in the same circle with positive power followed by the rotation of the said plane about the center of inversion by π radians, becomes especially evident.

Indeed. A rotation of a straight line in the parent plane by π radians about a point not on that straight line will take that line into a line that is **parallel** to the original, which precisely means that the magnitudes of the relevant angles under such a rotation will not change and the proofs of the above cases for positive power of inversion can be restated pretty much verbatim when the power of inversion is negative.

Theorem 34. Under the inversion of a plane with respect to a circle, $c(O, r)$, the magnitude of the angle between any two straight lines, l_1 and l_2 , none of which pass through the center of inversion, O , is equal to the magnitude of the angle between their inverse images, l'_1 and l'_2 .

Proof. Again, looking at an inversion with positive power, we consider three cases:

- the straight lines l_1 and l_2 are parallel
- the straight lines l_1 and l_2 are perpendicular
- the straight lines l_1 and l_2 are neither parallel nor perpendicular

In Chapter 5, across the Theorems 10, 11, 12, 13 and 14, we already proved that any straight line that does not pass through the center of inversion is taken by the corresponding inversion into a circle that does pass through the center of inversion.

Since in this theorem we have two straight lines neither of which passes through the center of inversion, it follows that the images of these straight lines will be the genuine circles each of which does pass through the center of inversion.

Moreover, if two circles intersect then they do so at two distinct points. Hence, the images of the straight lines in this theorem will intersect at two distinct points as well. One such point will, evidently, be the center of inversion, O , since both image circles must pass through it.

Let P be the second point of intersection of the images l'_1 and l'_2 of the given straight lines l_1 and l_2 .

Let t_1 and t'_1 be the tangent straight lines drawn to the circle l'_1 at the points O and P correspondingly.

Let t_2 and t'_2 be the tangent straight lines drawn to the circle l'_2 at the points O and P correspondingly.

Case 1. The straight lines l_1 and l_2 are parallel.

When the straight lines l_1 and l_2 are parallel then the angle between them is zero radians.

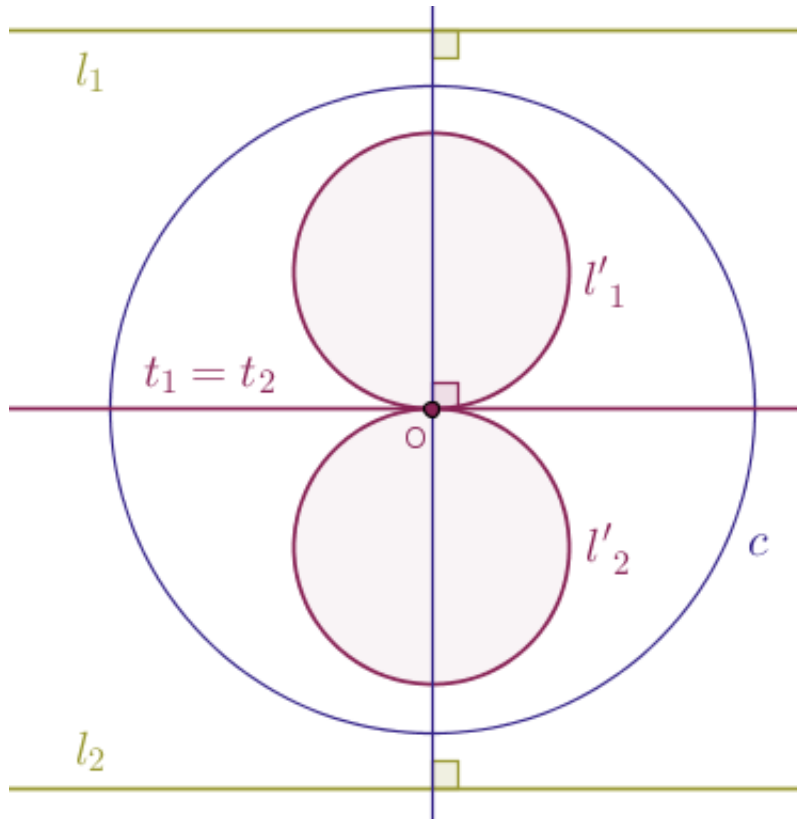
By the argument shown in Case 1 of the Theorem 33, it follows that in that case the straight line t_1 will be parallel to the straight line l_1 , or $t_1 \parallel l_1$, and that the straight line t_2 will be parallel to the straight line l_2 , or $t_2 \parallel l_2$.

By hypothesis, the straight lines l_1 and l_2 are parallel: $l_1 \parallel l_2$. In his [“Elements” Book 1 Proposition 30](#) Euclid proved that if two straight lines a and b are parallel to a third straight line, c , then these straight line are parallel among themselves: if $a \parallel c$ and $b \parallel c$ then $a \parallel b$.

Since in our case $t_1 \parallel l_1$ and $l_2 \parallel l_1$, it follows that $t_1 \parallel l_2$. Applying this argument to the tangent t_2 , we have.

Since $t_1 \parallel l_2$ and $t_2 \parallel l_2$, it follows that $t_1 \parallel t_2$.

Below we show a sample configuration that depicts this case when the center of inversion, O , is located in between the parallel preimage straight lines l_1 and l_2 (Figure 8.4):



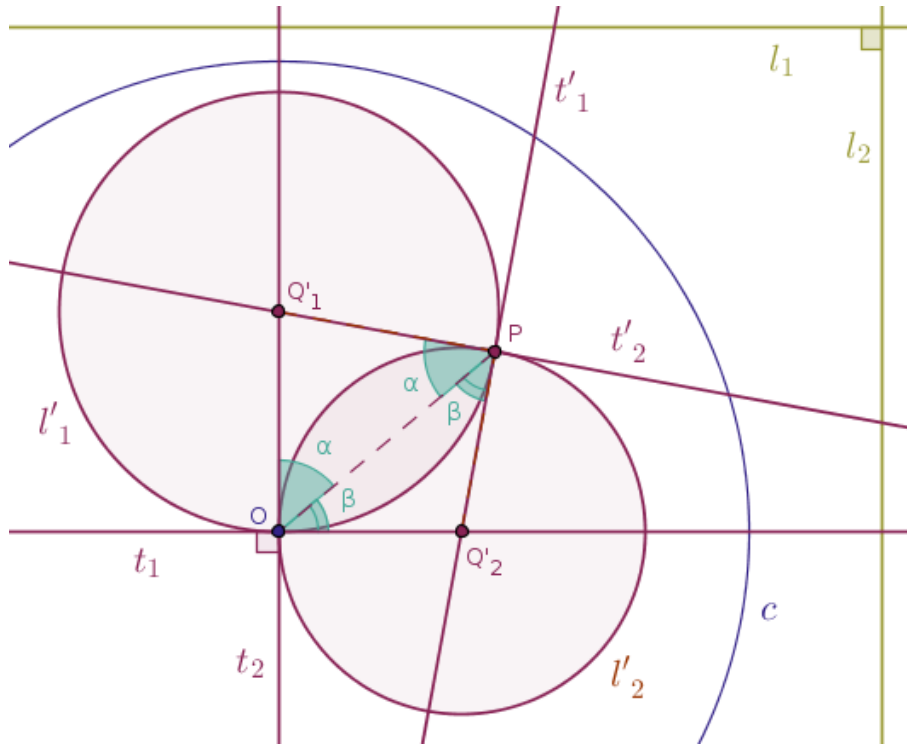
In that case the image circles l'_1 and l'_2 touch each other **externally**. It should be evident that when the preimage straight lines are located on the same side of the center of inversion then their images will touch each other **internally** but fundamentally the conclusion of this case will remain one and the same nonetheless.

Either way, since the tangents t_1 and t_2 both pass through one and the same point, the center of inversion, O , by the same Playfair's axiom, it follows that these straight lines coincide, making the angle between them also zero degrees. Therefore, in this case the magnitude of the angles between the original straight lines, l_1 and l_2 , and between their inverse images, l'_1 and l'_2 , is preserved. $\square$

Case 2. The straight line l_1 and l_2 are perpendicular.

When the straight lines l_1 and l_2 are perpendicular then by the same parallel postulate these two straight lines will intersect, forming an angle equal to $\pi/2$ radians.

By the same Book 1 Proposition 27 argument, the straight line t_1 will be parallel to the straight line l_1 , while the straight line t_2 will be parallel to the straight line l_2 . Thus, these four straight lines $l_{1,2}$ and $t_{1,2}$ will form a *parallelogram*. But by [Euclid's "Elements" Book 1 Proposition 34](#), since in such a parallelogram the opposite angles must be equal, the magnitude of the angle between the straight lines t_1 and t_2 , whatever it actually is, must be equal to the magnitude of the angle between the straight lines l_1 and l_2 (Figure 8.5):



which means that the magnitude of the angle between the original straight lines, l_1 and l_2 , and between their inverse images, l'_1 and l'_2 , at the point O is preserved.

Also, in the Theorem 33 we already proved that the triangles OQ'_1P and OQ'_2P will be isosceles. Hence, the following identity:

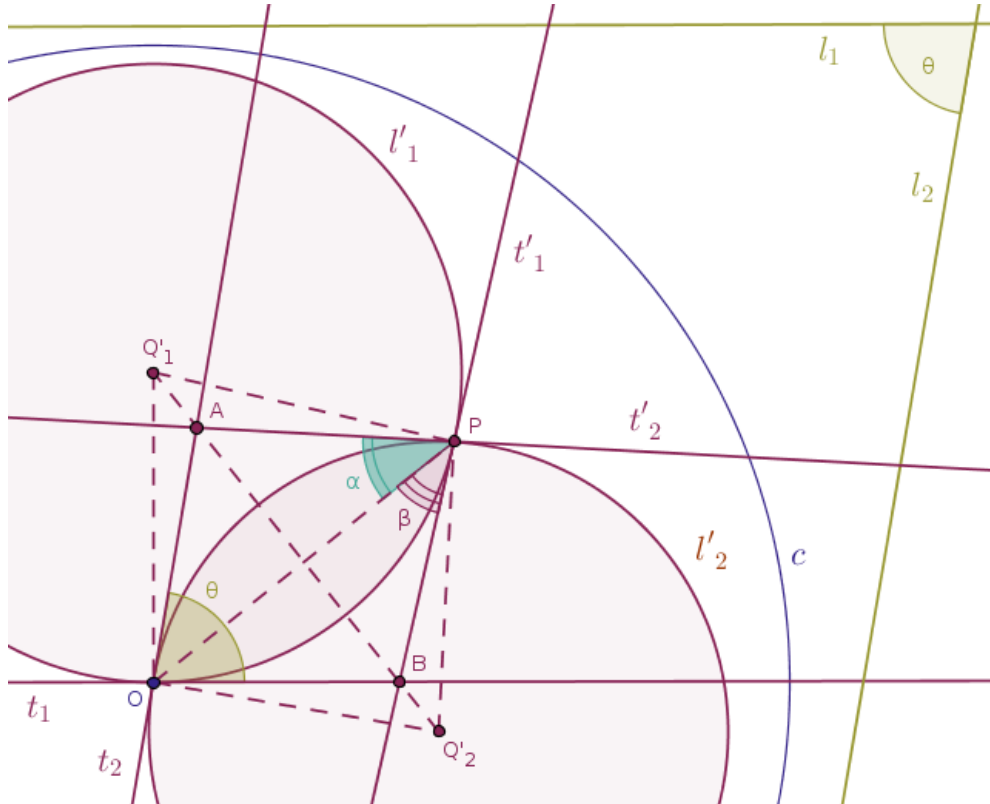
$$\angle Q'_1 O Q'_2 = \frac{\pi}{2} = \alpha + \beta$$

must hold at both vertices O and P alike. Therefore, the magnitude of the angle between the straight lines t'_1 and t'_2 will be equal to the magnitude of the angle between the straight lines t_1 and t_2 .

Thus, the magnitude of the angles between the original straight lines, l_1 and l_2 , and between their inverse images, l'_1 and l'_2 , at the point P is preserved also. $\square$

Case 3. The straight line l_1 and l_2 are neither parallel nor perpendicular.

When the straight lines l_1 and l_2 are neither parallel nor perpendicular then by the same parallel postulate these two straight lines will intersect. We, then, take the angle θ between these straight lines to be the smallest one (Figure 8.6):



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By the above Case 2 parallelogrammatic argument, the angle between the straight lines t_1 and t_2 will be equal to the angle between the straight lines l_1 and l_2 , which means that the magnitude of the angles between the original straight lines, l_1 and l_2 , and between their inverse images, l'_1 and l'_2 , at the point O is preserved.

However, this time the other two tangent straight lines, t'_1 and t'_2 , will not intersect their buddies, t_1 and t_2 , at the centers of the image circles, l'_1 and l'_2 .

Let t_1 intersect t'_1 at a point B and let t_2 intersect t'_2 at a point A and let the points Q'_1 and Q'_2 be the centers of the image circles l'_1 and l'_2 respectively.

Still, by construction, the triangles Q'_1OB and Q'_1PB are right. These two right triangles share a common hypotenuse Q'_1B and have two sides, Q'_1O and Q'_1P , of equal length, since both of these line segments are the radii of one and the same circle, l'_1 . By the Pythagorean theorem, then, it is the case that $|BO| = |BP|$, implying that the triangle OBP is isosceles and, hence, has the magnitudes of its interior angles at the base OP equal one another:

$$\angle BOP = \angle BPO = \beta$$

In exactly the same manner, the triangle OAP is isosceles and, hence, has the magnitudes of its interior angles at the base OP equal one another:

$$\angle AOP = \angle APO = \alpha$$

Since these identities must hold for both isosceles triangles at both vertices O and P , it follows that:

$$\angle AOB = \theta = \angle AOP + \angle BOP = \alpha + \beta = \angle APO + \angle BPO$$

implying that the magnitude of the angle between the straight lines t'_1 and t'_2 will be equal to the magnitude of the angle between the straight lines t_1 and t_2 .

Thus, the magnitude of the angle between the original straight lines l_1 and l_2 and the magnitude of the angle between their inverse images l'_1 and l'_2 at the point P is preserved also. $\square$

Our readers are encouraged to at least sketch the relevant diagrams for the above theorem if the power of inversion is negative and go, perhaps in their minds, over the steps of the corresponding proof on their own using the above proof as a guiding template.

As we have noted earlier, from the Definition 4 two more theorems follow quite mechanically because of the way in which we defined the angle between any two (cooperating) curves. That is.

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The angle between any two circles that have a point, T , in common is the angle between the tangent straight lines drawn to both of these circles through the said common point, T .

Precisely because these “tangents” are “straight lines” and in the above theorems we have covered all the possible cases in which inversion of a plane in a circle can act on the angle between any two straight lines, the following two theorems hold true.

Theorem 35. Under the inversion of a plane with respect to a circle, $c(O, r)$, the magnitude of the angle between a straight line, l , and a circle, q , is equal to the magnitude of the angle between their inverse images, l' and q' .

Theorem 36. Under the inversion of a plane with respect to a circle, $c(O, r)$, the magnitude of the angle between any two circles, q_1 and q_2 , is equal to the magnitude of the angle between their inverse images, q'_1 and q'_2 .

While we omit the proofs of these theorems, the reader is encouraged to construct them for practice nonetheless.